STAT4C03/6C03 Generalized Linear Models

Assignment 2 Due at 10pm on Thursday February 12, 2026

Instructions:

1. The only way to submit your assignment is through the Crowdmark page for the course. Use the link on Avenue (Content > Crowdmark) to login to Crowdmark and follow the instructions there to upload your assignment.

2. It is your responsibility to ensure that your submitted solutions are readable and complete. Any submitted file which cannot be opened will be given a mark of 0.

3. You may use R Markdown or Knitr to compile the R questions in your assignment but are not required to do so. For any question using R, you are required to include both the R code and output in your solutions as well as your written comments on the model(s) fitted.

4. Late assignments will be automatically assessed a penalty of 2% per hour, or part thereof, that they are late according to the Crowdmark time stamp.

5. You may update your submission up until the deadline. Once the due date has passed, you will no longer be able to update any submission. This includes adding questions that you did not submit before the deadline.

6. Please indicate **clearly** on your solutions whether you are in STATS 4C03 or STATS 6C03.

7. You are expected to show all details of your solution and any results taken from my notes or the textbooks must be clearly and properly referenced.

8. Students with accommodations through Student Accessibility Services who require accommodations for this assignment must contact the professor **at least 48 hours before the assignment is due** to request any such accommodations.

9. No other extensions to the due date and time will be given except in extreme circumstances.

10. There are 100 marks available for STATS 4C03 students and 125 for STATS 6C03 students. Note that the questions are not equally weighted. Marks for each question are given at the start of the question.

Q. 1 [30 marks]

In class we examined using the Newton-Raphson method to find the estimates of the parameters in a simple logistic regression for the Shuttle O-Ring data (you can get the data in the file `Shuttle.csv`). In this question you will examine the same data but do a probit analysis.

a) Derive the likelihood equations for β_0 and β_1 when using the probit link function.

b) Derive the matrix of second derivatives of the log-likelihood.

c) Suppose that $\beta_1 = 0$, derive the corresponding estimate $\hat{\beta}_0^{(0)}$ of β_0.

d) Implement the Newton-Raphson algorithm in R for this model giving your code as well as the estimates and the values of the log-likelihood derivatives at each iteration. Use the vector $(\hat{\beta}_0^{(0)}, 0)$ as the initial values to start the algorithm.

e) Give the asymptotic standard errors of the estimators.

Q. 2 [40 marks]

In a study of diabetes, the National Institute of Diabetes and Digestive and Kidney Diseases conducted a study on 768 adult females from the Pima native Americans (now known as the *Akimel O'odham* which translates to *River People*) living near Phoenix, Arizona. The dataset in `Pima.csv` contains this data. The response variable (`diabetes`) is the result of a diabetes diagnostic test. The covariates are:

pregnant Number of times pregnant

glucose Plasma glucose concentration at 2 hours in an oral glucose tolerance test

diastolic Diastolic blood pressure (mm Hg)

triceps Triceps skin fold thickness (mm)

insulin 2-Hour serum insulin (mu U/ml)

bmi Body mass index (weight in kg/(height in metres squared))

pedigree Diabetes pedigree function, a measure of family history of diabetes

age Age (years)

a) Fit a logistic regression model using all of the covariates and write a short report on your analysis.

b) For each of the covariates, compare the distribution for those who test positive and those who test negative. Do you notice anything strange for any of the covariates?

c) Replace the incorrect data (zeroes for variables which cannot be zero) with the missing value code `NA`. How many individuals have no missing values at all?

d) Refit the logistic regression model from (a) and comment on any differences.

e) Check if whether or not a person was included in the analysis in (d) is associated with the response. That is, is there a different proportion having diabetes depending on whether or not individuals were included in the model? What does this result mean for our analysis?

f) Fit a model without the insulin and triceps predictors and do a likelihood ratio test to compare this reduced model with the full model. Remember that your reduced model must be fitted on **exactly the same data** as the full model for the likelihoods to be comparable. If the null hypothesis is not rejected, fit the reduced model using as much data as possible. How many data points can you now use?

g) Use a two-sample t-test to compare the mean diastolic blood pressure for those who test positive and those who test negative for diabetes. Compare the result to a test of the significance of diastolic blood pressure in your simplest logistic regression. Can you give a reason why these two tests do not agree?

Q. 3 [30 marks]

The dataset in `Fastener.csv` gives data on the failures of alloy fasteners used in aircraft manufacturing when subjected to high pressure loads. The three columns are the pressure used, the number of fasteners tested and the number which failed.

a) Fit a logistic regression model using pressure as the only covariate and write a paragraph to explain the results of your analysis.

b) Give a table of the fitted failure probabilities from both the saturated model and the logistic model from part (a).

c) The failure probability curve is a plot of the fitted probabilities against the covariate with each successive point connected with a line. On the same plot, show the failure probability curves for the saturated and the fitted logistic regression model. Comment on the differences between the two curves.

d) Calculate the residual deviance using the formula on Page 3-30 of my notes and verify that the value returned by the R `summary` agrees with your calculations.

e) For this dataset find the test statistic and corresponding asymptotic p-value for the Pearson's goodness of fit test. How do these compare to the deviance goodness of fit test.

f) Conduct a likelihood ratio test comparing the fitted model to the model with no covariate. Make sure to give the value of the test statistic, specify the asymptotic distribution of the test statistic if pressure has no effect on the failure probability, give the corresponding p-value and clearly state your conclusions.

Q. 4 STATS 6C03 ONLY **[25 marks]** We often deal with 2×2 contingency tables where we compare a binary outcome with a binary covariate. Consider, for example, seeing if person having Coronary Artery Disease (CAD) at age 50 is related to the presence or absence of a specific genetic mutation in their DNA (fixed at conception). This results in a table such as

	Has CAD	No CAD
Mutation	π_1	$1 - \pi_1$
No Mutation	π_2	$1 - \pi_2$

where π_1 and π_2 are the probabilities of having CAD for people with and without the mutation respectively. The odds ratio is defined as

$$\phi \;=\; \left(\frac{\pi_1}{1-\pi_1}\right) / \left(\frac{\pi_2}{1-\pi_2}\right) \;=\; \frac{\pi_1(1-\pi_2)}{\pi_2(1-\pi_1)}$$

a) We could fit a simple logistic model $\pi_i \;=\; \mathrm{e}^{\beta_i}/(1+\mathrm{e}^{\beta}_i)$ for $i = 1, 2$. Show that if $\beta_1 = \beta_2$ then $\phi = 1$.

b) Now suppose that there is another ordered factor X with J equally spaced levels $x_1,] \ldots, x_J$ and, at each value of X, we have a 2×2 table as above. Let ϕ_j be the odds ratio for the two binary factors when $X = x_j$. Suppose we fit the logistic model

$$pi_{ij} \;=\; \frac{\mathrm{e}^{\alpha_i + \beta_i x_j}}{1 + \mathrm{e}^{\alpha_i + \beta_i x_j}} \qquad \text{for } i = 1, 2; \quad j = 1, \ldots, J$$

Show that all of the ϕ_j are equal when $\beta_1 = \beta_2$. What is the constant that they are equal to if the disease is not related to either of the covariates?